

# TEZPUR UNIVERSITY

## Spring Semester MAJOR I, 2018

*Design and Analysis of Algorithms* (CS 601/ CS 701)

Date : March 20, 2018

Maximum Marks : 40  
minutes

Duration : 75

**You may answer all the questions but maximum you can get is 40.**

1. For each of the following statements indicate whether the statement is true or false and briefly explain why.

a) If  $T(N) = 2T(\frac{N}{2}) + 1$  and  $T(\frac{N}{100}) = 1$ , then  $T(N) \in \Theta(N \log N)$ . [3]

b)  $(\log(N))^2 \in O(\log(N))$  [3]

c) For any  $n \geq 0$ ,  $n(n^2 - 1)$  is divisible by 3. [5]

d) It takes  $O(\log n)$  time to find the rank of an element in a unsorted array. [3]

e) The maximum number of comparisons required to find an element in a sorted array of 25 elements is 6. [3]

f)  $f(cn) \in \Theta(f(n))$ . [3]

g) A binary tree with 16 leaf nodes cannot have a height greater than 5. [3]

h) There exists an array  $A$  of  $n$  elements, that can be sorted in  $O(n)$  time using INSERTION SORT algorithm. [3]

i) The function call `recurse("samamas", 0, 6)` will return TRUE. [4]

```
int recurse(char *s, int a, int b) {  
    if (a < b){  
        if (s[a] == s[b])  
            return recurse(s, a + 1, b - 1);  
        else  
            return FALSE;  
    }  
    else  
        return TRUE;  
}
```

2. Consider the linear time algorithm algorithm for finding the  $k$ th highest element in a array. How many recursive calls are required to find the 4th highest element in the array? [7]

$< 24, 16, 21, 6, 8, 19, 20, 5, 7, 4, 9, -5 >$

.

3. How many times distance between two points is computed by the closest-pair algorithm for the following two case? [10]

The points are at locations (0,0), (1,1), (2,2), (3,3), (4,4), (5,5), (6,6), (7,7).

4. Demonstrate the divide-and-conquer multiplication algorithm by computing  $1110 \times 1011$ . You don't need to write the algorithm, just show how your algorithm works on the above input. [10]